

PROTOSPHERE - Embedded Systems & IoT Bootcamp

Duration: 7 Hours (Across 2 Days)

Participants will learn how hardware and software come together to create intelligent systems, understand how to select the right components for a project, interface sensors and actuators, and program microcontroller specifically ESP32.

Instead of simply following instructions to build one project, students will gain the knowledge and confidence to design and develop their own embedded projects independently. Throughout the workshop, short hands-on activities and live demonstrations will reinforce each concept, making the learning process interactive and engaging

Components required -

- ESP32
- LCD (16x4)
- Ultrasonic sensor
- breadboards/wires etc

Event Flow

Day 1 -- Embedded Systems Fundamentals & Hardware Interfacing

Welcome & Workshop Overview

- Introduction to the workshop
- Learning objectives and expected outcomes

Embedded Systems & IoT Fundamentals

- What is an Embedded System?
- What is the Internet of Things (IoT)?
- Where does the ESP32 fit?
- Comparison with Arduino, STM32, PIC, and Raspberry Pi

ESP32 Hardware Overview

- High-level ESP32 architecture

- Development board walkthrough
- GPIOs, power, onboard components
- Pin overview and best practices

Communication Protocols

- UART
- SPI
- I²C
- How embedded devices communicate

Development Environment Setup

- Install Arduino IDE
- Install the ESP32 Board Package
- Configure the development environment
- Upload the first program to verify the setup

Sensors & Peripherals

- Ultrasonic sensor
- 16×4 LCD display
- Brief overview of peripheral interfacing

Hands-on Activity 1 -- Ultrasonic Sensor

- Interface the ultrasonic sensor
- Read distance values
- Implement simple distance-based logic

Hands-on Activity 2 -- LCD Display

- Interface the LCD
- Display custom text
- Display live ultrasonic sensor readings

Day 2 -- Wireless Communication & IoT

Introduction to IoT Networking

- Device-to-device communication
- Wi-Fi vs Bluetooth vs ESP-NOW

- Why use ESP-NOW?

Hands-on Activity 3 -- ESP-NOW Communication

- Create a wireless sensor node
- Send sensor values between ESP32 boards
- Display received values on a remote LCD

Hands-on Activity 4 -- ESP-NOW Applications

- Wireless messaging between participants
- Broadcast notice board
- Wireless quiz / voting demonstration

Wi-Fi & Embedded Web Server

- ESP32 Access Point mode
- Host a local webpage
- Connect using smartphones
- Control the LCD from a phone
- View live sensor data on a phone

IoT Gateway Demonstration

- Combine ESP-NOW and Wi-Fi
- Sensor Node → Gateway → Smartphone architecture
- Offline phone-to-phone communication demonstration

Introduction to FreeRTOS

- Why an RTOS?
- Tasks and scheduling
- FreeRTOS on the ESP32
- Simple multitasking demonstration

Mini Challenge

Build a small embedded or IoT application using the concepts learned during the workshop.

Wrap-up

- Project showcase

- Q&A session
 - Learning roadmap and next steps
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Key Takeaways

Participants will have progressed through the complete embedded systems development stack:

- Understanding embedded systems and IoT fundamentals.
- Exploring the ESP32 architecture and hardware.
- Learning communication protocols (UART, SPI, I²C).
- Interfacing sensors and displays.
- Building embedded applications with peripherals.
- Creating device-to-device communication using ESP-NOW.
- Developing smartphone-accessible web interfaces.
- Understanding gateway-based IoT architectures.
- Getting a first look at multitasking with FreeRTOS.